12.1 Displaying Univariate Data

Lesson Introduction

Benchmarks	MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data, and on whether it is univariate or bivariate. MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.	Learning Targets	• I can understand how various displays can be used to represent a univariate categorical data set. • I can select an appropriate data display to represent numerical or categorical univariate data.

Benchmark Coherence	Building On	Working Toward
	MA.8.DP.1.1 Given a set of real-world, bivariate numerical data, construct a scatter plot or a line graph as appropriate for the context.	N/A

Essential Questions	• How can various displays be used to represent a univariate categorical data set? • How can I determine the appropriate data display to represent numerical or categorical univariate data?
Lesson Narrative	In this lesson, students are refreshed on the concept of univariate data. Students will begin with a refresher on choosing the best graphical representation for univariate numerical data. They will then collaborate to graph representations of univariate categorical data.

Component Summary				
	12.1.1 Warm-Up (~5 min.)	Bell Work on categorical vs numerical data *(SE p. 195)*	12.1.3 Collaboration (15-20 min.)	Collaboration on graphical representations of univariate categorical data *(SE p. 199)*
	12.1.2 Refresher (15-20 min.)	Refresher on choosing the best graphical representation for univariate numerical data *(SE p. 196)*	12.1.4 Wrap-Up (~5 min.)	Share favorite graphical representations *(SE p. 201)*

Resources	Glossary	Materials	Additional Preparation
	Key Terms: • Univariate Data Additional Terms: N/A	• (Optional) Signaling Devices (i.e., Whiteboard Paddles)	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may choose the wrong scaling for a visual representation.
Student may misidentify categorical data that contains numbers (i.e., zip codes).</td><td>Students created and analyzed displays in 6th and 7th grades. See 6th Grade Unit 9 and 7th Grade Unit 13 for additional supporting material.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
12.1.3 Collaboration provides students with a collaborative opportunity with graphical representations of univariate categorical data. Students will have individual time with the activity, and then collaborative partner work in other components of the activity.</td><td>MA.K12.MTR.2.1: Multiple Representations
In this lesson, students review data representations displayed in different ways. Throughout the lesson, students will represent solutions to problems in multiple ways using objects, drawings, tables, graphs, and equations. They will express connections between concepts and representations.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider creating a visual reference guide for these different visual data representations. Each display should include the different scenarios where that would be the best display of the data.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

12.1.1 Warm-Up: Bell Work

Component Overview: *Students will determine whether each given scenario represents categorical or numerical data.*

12.1.1 Warm-Up: Bell Work

To choose an appropriate graphical representation for data, the first step is to determine if the data is ***numerical*** (data values are numbers) or ***categorical*** (data values fall within categories).

1. Complete the table.

Possible Question	Possible Responses	Numerical or Categorical
What is your favorite color?	red, orange, yellow, green	● categorical ○ numerical
How many people are in your family?	$2, 3, 4, 5 \ldots$	○ categorical ● numerical
How old are you?	$11, 12, 13 \ldots$	○ categorical ● numerical
How tall are you?	*Answers will vary.*	○ categorical ● numerical
What is your favorite sport?	*Answers will vary.*	● categorical ○ numerical
What is your favorite fruit?	watermelon, apple, pear, orange	● categorical ○ numerical

Consider adding a layer of engagement to the review of this activity. Provide students with whiteboard paddles, sheets of papers, or other signaling devices. Call out the question, and then have students raise their signaling device to share their answer.

12.1.2 Refresher: Choosing the Best Graphical Representation for Univariate Numerical Data

Component Overview: *Students will collaborate on choosing the best graphical representation for univariate numerical data by first analyzing attributes each representation provides. Then, students will be provided with a scenario where they will have to choose the best graphical representation for the data. This refresher can be extended with resources from 7th Grade Unit 13.*

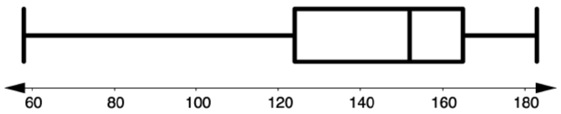
12.1.2 Refresher: Choosing the Best Graphical Representation for Univariate Numerical Data

Four graphical representations that can be used to display ***numerical data*** that is ***univariate*** are box plots, histograms, line plots, and stem-and-leaf plots. Each of these graphical representations is shown in the table.

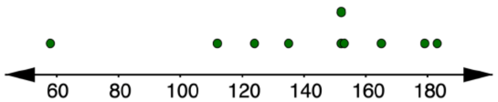
Univariate data is data that measures one characteristic of a population.

Graphical Representations for Univariate Quantitative Data

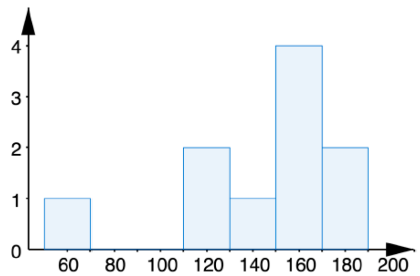
Box Plot



Line Plot



Histogram



Stem-and-Leaf Plot

Stem	Leaf
5	8
6	
7	
8	
9	
10	
11	2
12	4
13	5
14	
15	2, 2, 3
16	5
17	9
18	3

Key: 5|8 = 58

Notice that the words "numerical" and "quantitative" are used interchangeably. Make sure to point this out to students when they are analyzing the graphical representations for univariate quantitative data.

Consider creating a visual reference guide for these different visual data representations. Each display should include the different scenarios where that would be the best display of the data.

What does it mean when there is no value in the leaf column of a stem-and-leaf plot?

12.1.2 Refresher: Choosing the Best Graphical Representation for Univariate Numerical Data (continued)

Work with your partner to complete the following.

1. What do you notice about the displays?

   Answers will vary. I notice…There is a gap in the data from 58 to 112. There is a possible outlier of 58. The range of the data is 125. I wonder… What is this data from? Why is there such a large gap in the data?

2. Complete the table by checking the displays that match each attribute.

Attribute	Box Plot	Line Plot	Histogram	Stem and Leaf Plot
Shows the distribution of the data set	✓	✓	✓	✓
Shows the shape of the data distribution	✓	✓	✓	✓
Data is displayed in intervals (also called bins)	☐	☐	✓	✓
Displays each individual data value	☐	✓	☐	✓
Displays a 5-number summary of the data set	✓	☐	☐	☐

Have students share their answers for question 2 through a whole-group activity. Add a physical element by placing the name of each display type in four corners of the room. Call out an attribute and see which ones students go to. Elicit student responses to agree or disagree, and engage in discussion on which ones apply to which displays.

12.1.2 Refresher: Choosing the Best Graphical Representation for Univariate Numerical Data (continued)

3. Camilla wants to create a graphical display that represents the typing speeds of the 425 students at her school.

   a. Which display(s) will best display the data?

   - ☐ box plot
   - ☐ line plot
   - ☐ histogram
   - ☐ stem-and-leaf plot

   b. Explain your reasoning.
   Answers will vary. The box plot and histogram are best for this data because they show the overall shape of the data. Both the stem and leaf plot and the line plot could be hard to read and construct because of the large amount of data. In this case we don't need to see every point, so a box plot or histogram would be best.

   c. Camilla says, "Making a line plot of this data would take a lot of time and it would be hard to make sure each dot is in the right location on the number line, since speed can be given in fractions of seconds. I'd have to make 425 dots to represent each student!" Explain whether you agree with Camilla that a line plot would not best display the data.
   Answers will vary. I agree with Camilla. Another reason a line plot isn't so great for this data is the range is likely quite small which could make the line plot really tall. With this much data, a box plot or histogram would do a better job of showing the overall shape of the data in an efficient way.

Have students share out their answers to questions 3a and 3b. Leverage an agree/disagree protocol to have students justify their answer and whether they agree or disagree with other students answers.

Ideally, after reading question 3c, students will recognize which are appropriate and inappropriate for large data sets. They can go back to revise their answers to questions 3a and 3b.

What other displays would not be good for this data set? Why not?

12.1.2 Refresher: Choosing the Best Graphical Representation for Univariate Numerical Data (continued)

4. Dallas records the highest temperature from each month over the past year in Immokalee, Florida.

$$\{76, 78, 81, 85, 90, 92, 92, 93, 91, 87, 82, 78\}$$

a. Which display(s) will best display the data?

- ☐ box plot
- ☐ line plot
- ☐ histogram
- ☐ stem-and-leaf plot

b. Explain your reasoning.
Answers will vary. A line plot is a good way to display this data because there is a small amount of data, and each value will be visible. Plus, the scale of a line plot is more flexible than the scale of a stem-and-leaf plot, so a line plot may show the shape of the data better for this data set since it has got quite a small range. A box plot is another good way to display this data. It will show the overall shape of the data and the five number summary which will give a good picture of the data.

Have students share out their answers to questions 4a and 4b. Leverage an agree/disagree protocol to have students justify their answer and whether they agree or disagree with other students answers.

c. Dallas decides to change his data set to include the average weekly high temperature. Explain whether this changes the display he should use.
Answers will vary. Yes, with that much more data a more efficient way to display it would be a box plot. This will still give an overall picture as to the shape of the data, but it is a more condensed way to display the data.

Why would the change in data type result in a change in data display? Would that always be true? Why or why not?

12.1.2 Refresher: Choosing the Best Graphical Representation for Univariate Numerical Data (continued)

5. Basil recorded the weights, rounded to the nearest pound, of 15 dogs at a pet show.

   $$7, 10, 12, 22, 28, 59, 62, 65, 72, 79, 85, 98, 100, 150, 163$$

   a. Explain which display would be better for the data, a line plot or a stem-and-leaf plot.
   Answers will vary. A stem-and-leaf plot is good for this data set because it allows the data to be grouped by tens. A stem-and-leaf plot will also show each data value and make gaps and clusters easier to see.

   b. Basil also recorded the weights, rounded to the nearest pound, of 15 cats at the pet show.

   $$4, 4, 5, 7, 7, 8, 8, 8, 9, 10, 11, 12, 13, 13, 14$$

   Explain which display would be better for the data, a line plot or a histogram.
   Answers will vary. In this case because the range of the data is small and there are not that many data points, a line plot is best.

   c. He updates his data to show the weights to the nearest tenth.

   $$4.1, 4.3, 5.2, 7.6, 7.9, 8.0, 8.2, 8.7, 9.3, 10.5, 11.2, 12.8, 13.4, 13.5, 14.6$$

   Explain which display would be better for the data, a line plot or a histogram. If your recommendation is different from Part B, explain why.
   If the data is rounded to the nearest tenth of a pound, the best display for it will be a histogram. With a histogram the data can be grouped into intervals which makes labeling the x-axis much more simple.

What is the role of rounding and its impact on the type of visual display that is chosen?

12.1.2 Refresher: Choosing the Best Graphical Representation for Univariate Numerical Data (continued)

6. Ms. Rallen asked her students how many pets they each have, and wants to use a graphical representation that will allow her students to see the overall shape of the data and give them enough information to calculate the 5-number summary.

   a. Which graphical representations should Ms. Rallen consider?

      - ☐ box plot
      - ☐ line plot
      - ☐ histogram
      - ☐ stem-and-leaf plot

   b. Explain your reasoning.
      Answers will vary. All of these representations, box plot, line plot, and stem-and-leaf plot will give information that can be used to calculate the five number summary. However, a box plot already has many of the numbers in the five number summary calculated, so if Ms. Rallen wants students to calculate as many of the numbers in the five number summary as possible, she probabably shouldn't use the box plot.

At the conclusion of 12.1.2 Refresher, consider engaging the class in a whole-group discussion on the learning from this activity. Ask students to share the strengths and weaknesses of each visual representation.

12.1.3 Collaboration: Graphical Representations of Univariate Categorical Data

Component Overview: *Students will collaborate on graphical representations of univariate categorical data using four practice questions. For 12.1.3 Collaboration, students will need a partner.*

12.1.3 Collaboration: Graphical Representations of Univariate Categorical Data

Graphical representations that can be used to display univariate categorical data include ***bar graphs***, ***circle graphs, line plots, frequency tables,*** and ***relative frequency tables.***

Categorical data is often collected in a frequency table before creating another display.

Consider the data summarized in the frequency table and relative frequency table, collected from a survey on how students get to school.

Frequency Table

Mode of Transportation	Frequency
Bus	129
Car	71
Bike	65
Walk	35

Relative Frequency Table

Mode of Transportation	Relative Frequency
Bus	43%
Car	24%
Bike	22%
Walk	11%

How can this table be read? What's an example of information we can determine from this table?

1. Summarize the similarities and differences between the two displays.

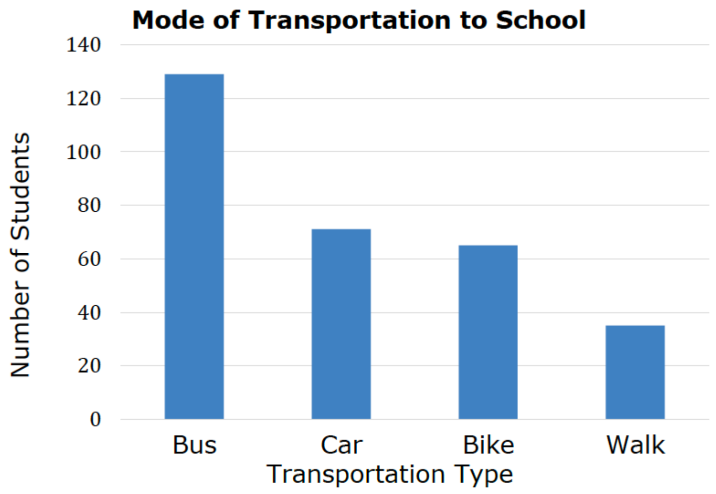
Similarities	Differences
Answers will vary. Both displays have the same categories in the same order. The displays are made using the same data. In both displays the labels are on the left and the numbers are on the right. Both displays show a frequency, one is a relative frequency. Both displays are in a table format.	*Answers will vary. The frequency table shows the data as integers while the relative frequency table shows the data as percentages. In the frequency table it is clear exactly how many people surveyed travel by bus, car, bike, walk, and it's easy to find the total people surveyed. In the relative frequency table, it is not possible to know how many people were surveyed with the information given.*

2. How was the data from the frequency table used to construct the relative frequency table?
Answers will vary. To calculate the percentages in the relative frequency table, the total of the data in the frequency table was found. Then this number was used as the whole in ratios to find the percentages in the relative frequency table. Here is an example:
$\frac{129}{300} = \frac{43}{100} = 43\%$

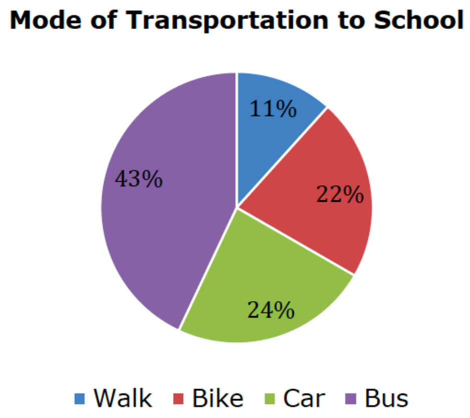
Consider creating a large T-chart to put up the answers that you observe while circulating the room. When 12.1.3 Collaboration ends, refer to this chart to highlight things that were heard and observed during 12.1.3 Collaboration as part of the discussion.

12.1.3 Collaboration: Graphical Representations of Univariate Categorical Data (continued)

The vertical bar graph and circle graph show the same data.



The circle graph shows the same data.



Which representation do you think better represents the data? Why?

3. Complete the statements.

   The bar graph visually shows

   - ○ how each part relates to the whole.
   - ● how each part compares to the other parts.

   The circle graph shows

   - ● how each part relates to the whole.
   - ○ how each part compares to the other parts.

   The data from the frequency table was used to construct the

   - ● bar graph.
   - ○ circle graph.

   The data from the relative frequency table was used to construct the

   - ○ bar graph.
   - ● circle graph.

4. Similar to a line plot (dot plot) for quantitative data, a line plot can also be used for categorical data by listing each category on the x-axis. Would a line plot be a good choice for displaying this data? Explain your reasoning.
   Answers will vary. A line plot would not be a good choice for displaying this data. Even though a line plot is very similar to a bar graph, line plots don't work well with large amounts of data. Since there was a large number of people surveyed, and over 100 of them fall into one category alone, a line plot for this data would take up way too much space.

Students may struggle with the sentence construction. Listen for students getting started, and provide support on how to approach these problems if they are struggling. For example, have them write the sentence on a separate sheet of paper to help them visualize the sentence to see if their choices create an accurate statement.

Which data display closely resembles the visual representation of a line plot?

12.1.4 Wrap-Up: Cool Down

Component Overview: *Students will share which graphical representations they prefer for displaying univariate categorical and numerical data and why.*

12.1.4 Wrap-Up: Cool Down

1. My favorite graphical representation to display

   numerical univariate data is
   - box plot.
   - line plot.
   - histogram.
   - stem and leaf.

   It is my favorite because ...

   Answers will vary.

2. My favorite graphical representation to display

   univariate categorical data is a
   - bar graph.
   - circle graph.
   - line plot.
   - frequency table.

   It is my favorite because ...

   Answers will vary.